

Física Computacional

ano letivo: 2021/2022

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Licenciatura em Física e Licenciatura / Mestrado Integrado em Engenharia Física

3ª aula (21/10/2021)



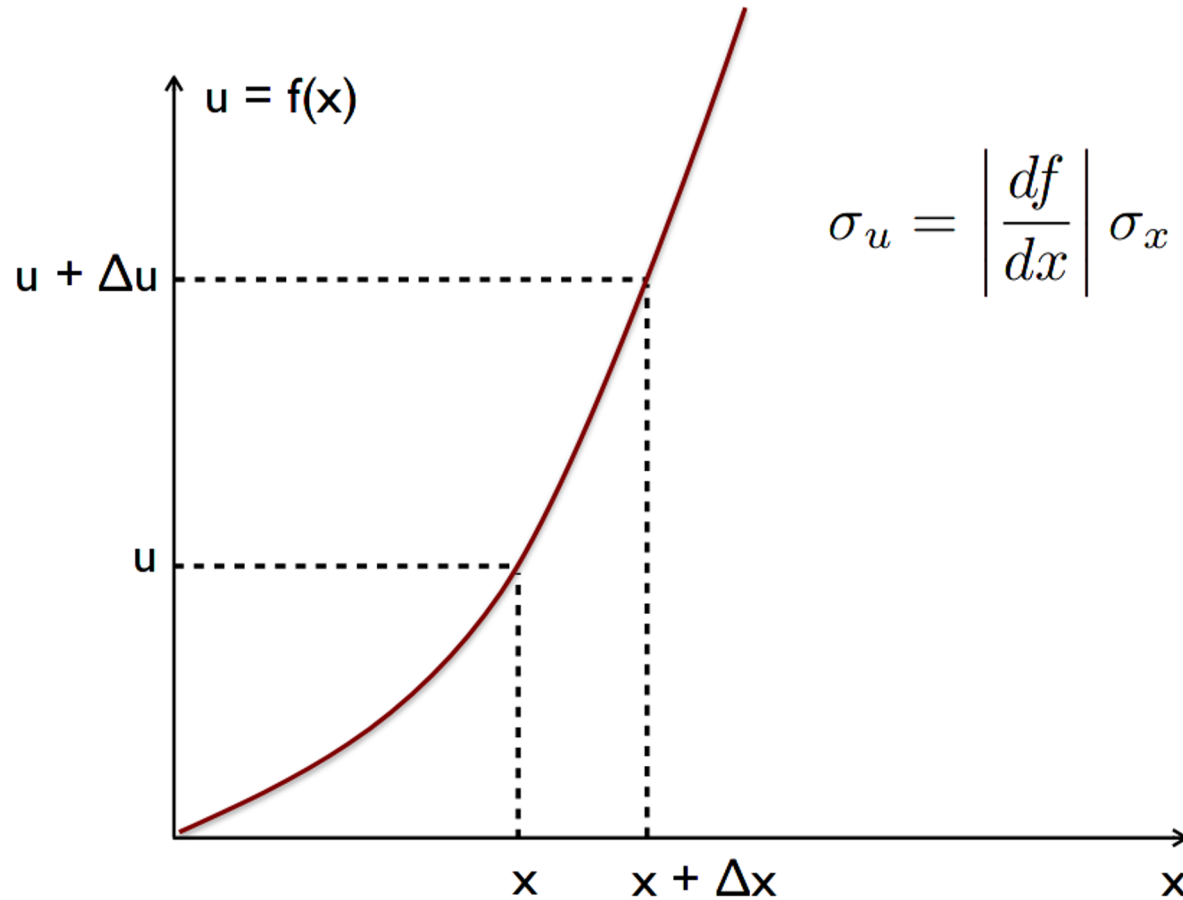
Lei de Gauss sobre a propagação do erro

Sejam $x_1, x_2, \dots, x_m$ variáveis e $\Delta x_1, \Delta x_2 \dots \Delta x_m$ os seus erros.
Seja $f(x_1, x_2, \dots, x_m)$ uma função qualquer destas variáveis.

Então o erro de f pode ser estimado por

$$\Delta f \simeq \sqrt{\sum_{j=1}^m \left(\frac{\partial f}{\partial x_j} \Delta x_j \right)^2}$$

Propagação de erros



Propagação de erros

Seja $x = 2.34(7)$ e $y = 5.689(5)$.

Qual será o erro da expressão $f(x, y) = x^2 y^3$?

$$x = 2.34(7) \Rightarrow x = 2.34 \pm 0.07$$

$$y = 5.689(5) \Rightarrow y = 5.689 \pm 0.005$$

Então $\Delta x = 0.07$ e $\Delta y = 0.005$. Assim:

$$\Delta f \simeq \sqrt{\sum_{j=1}^m \left(\frac{\partial f}{\partial x_j} \Delta x_j \right)^2} = \sqrt{\left(\frac{\partial f}{\partial x} \Delta x \right)^2 + \left(\frac{\partial f}{\partial y} \Delta y \right)^2}$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} x^2 y^3 = 2xy^3$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} x^2 y^3 = 3x^2 y^2$$

Propagação de erros

Então

$$\begin{aligned}\Delta f &\simeq \sqrt{\left(\frac{\partial f}{\partial x}\Delta x\right)^2 + \left(\frac{\partial f}{\partial y}\Delta y\right)^2} \\&= \sqrt{(2xy^3\Delta x)^2 + (3x^2y^2\Delta y)^2} \\&= \sqrt{(2 \cdot 2.34 \cdot 5.689^3 \cdot 0.07)^2 + (3 \cdot 2.34^2 \cdot 5.689^2 \cdot 0.005)^2} \\&\simeq \sqrt{60.318661309^2 + 2.6582439946^2} \\&\simeq \sqrt{3645.4071633} \simeq 60.377207316 \simeq 60\end{aligned}$$

Propagação de erros

Resultado:

$$f(x, y) = x^2 y^3 \simeq 1008 \pm 60 = (1.008 \pm 0.060) \cdot 10^3$$

Notação correta:

$$f(x, y) = (1.01 \pm 0.06) \cdot 10^3$$

$$f(x, y) = 1.01(6) \cdot 10^3$$

Se olharmos para os cálculos, nota-se que o erro final é dominado pelo erro em x (na soma dos quadrados por baixo da raiz a parte correspondente ao x ($\simeq 60.3$) é muito maior do que a parte correspondente ao y ($\simeq 2.7$).

Isto é assim porque o erro relativo de x é muito maior do que o erro relativo de y :

$$\frac{\Delta x}{x} = \frac{0.07}{2.34} = 0.030 = 3.0\% \qquad \frac{\Delta y}{y} = \frac{0.005}{5.689} = 0.00088 = 0.088\%$$

Em geral: $u = f(x, y)$

$$\sigma_u^2 = \left(\frac{\partial f}{\partial x} \right)^2 \bigg|_{(\bar{x}, \bar{y})} \sigma_{\bar{x}}^2 + \left(\frac{\partial f}{\partial y} \right)^2 \bigg|_{(\bar{x}, \bar{y})} \sigma_{\bar{y}}^2 + \frac{2}{N} \left(\frac{\partial f}{\partial x} \right) \left(\frac{\partial f}{\partial y} \right) \bigg|_{(\bar{x}, \bar{y})} \sigma_{xy}$$

$$\bar{u} = f(\bar{x}, \bar{y})$$

$$\text{i) } u = x \pm y \quad \longrightarrow \quad \sigma_{\bar{u}} = \sqrt{\sigma_{\bar{x}}^2 + \sigma_{\bar{y}}^2 \pm 2r\sigma_{\bar{x}}\sigma_{\bar{y}}}$$

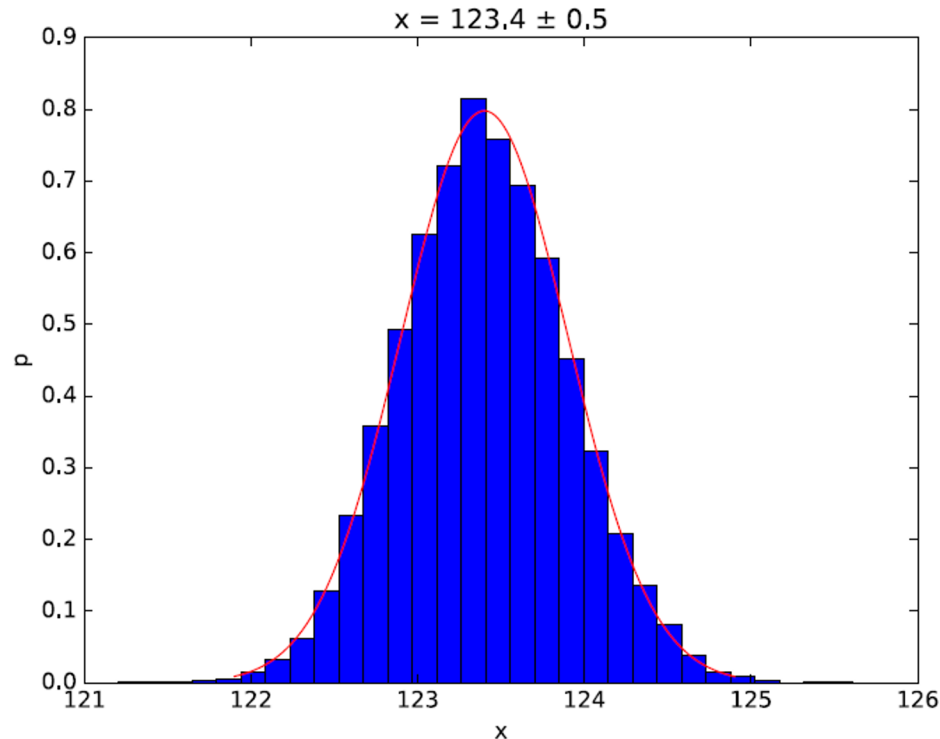
$$\begin{array}{l} \text{ii) } u = xy \\ \quad \text{ou} \\ u = x/y \end{array} \quad \longrightarrow \quad \frac{\sigma_{\bar{u}}}{|\bar{u}|} = \sqrt{\left(\frac{\sigma_{\bar{x}}}{\bar{x}} \right)^2 + \left(\frac{\sigma_{\bar{y}}}{\bar{y}} \right)^2 \pm 2r \left(\frac{\sigma_{\bar{x}}}{\bar{x}} \right) \left(\frac{\sigma_{\bar{y}}}{\bar{y}} \right)}$$

O significado do erro

- Exemplo: $x = 123.4 \pm 0.5$
- O que significa “ ± 0.5 ” ?
- Imaginamos que íamos medir o valor um número infinito de vezes utilizando o mesmo procedimento.
- Será que significa que o valor medido está sempre no intervalo $[123.4 - 0.5, 123.4 + 0.5]$?

O significado do erro

- Fazemos a “experiência” ...
- “medimos” o valor n vezes, $n = 10000$.
- Vamos obter o seguinte resultado da distribuição dos valores obtidos:

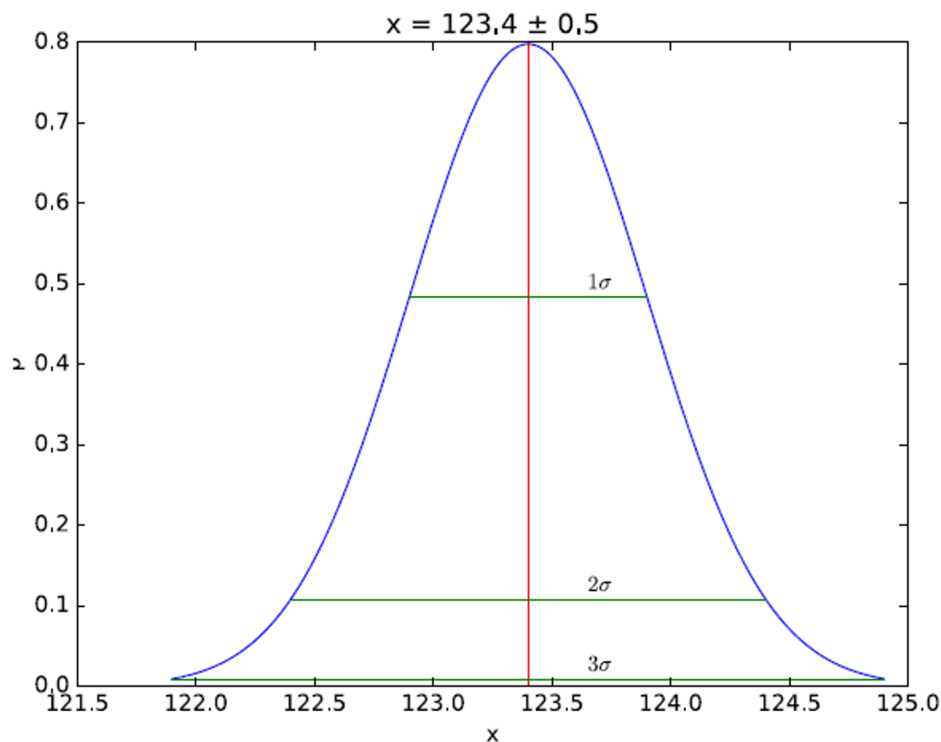


O significado do erro

- Na aproximação de $n \rightarrow \infty$ obtemos uma **Distribuição normal** ou **Gaussiana**:

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

com $\mu = 123.4$ e $\sigma = 0.5$



O significado do erro

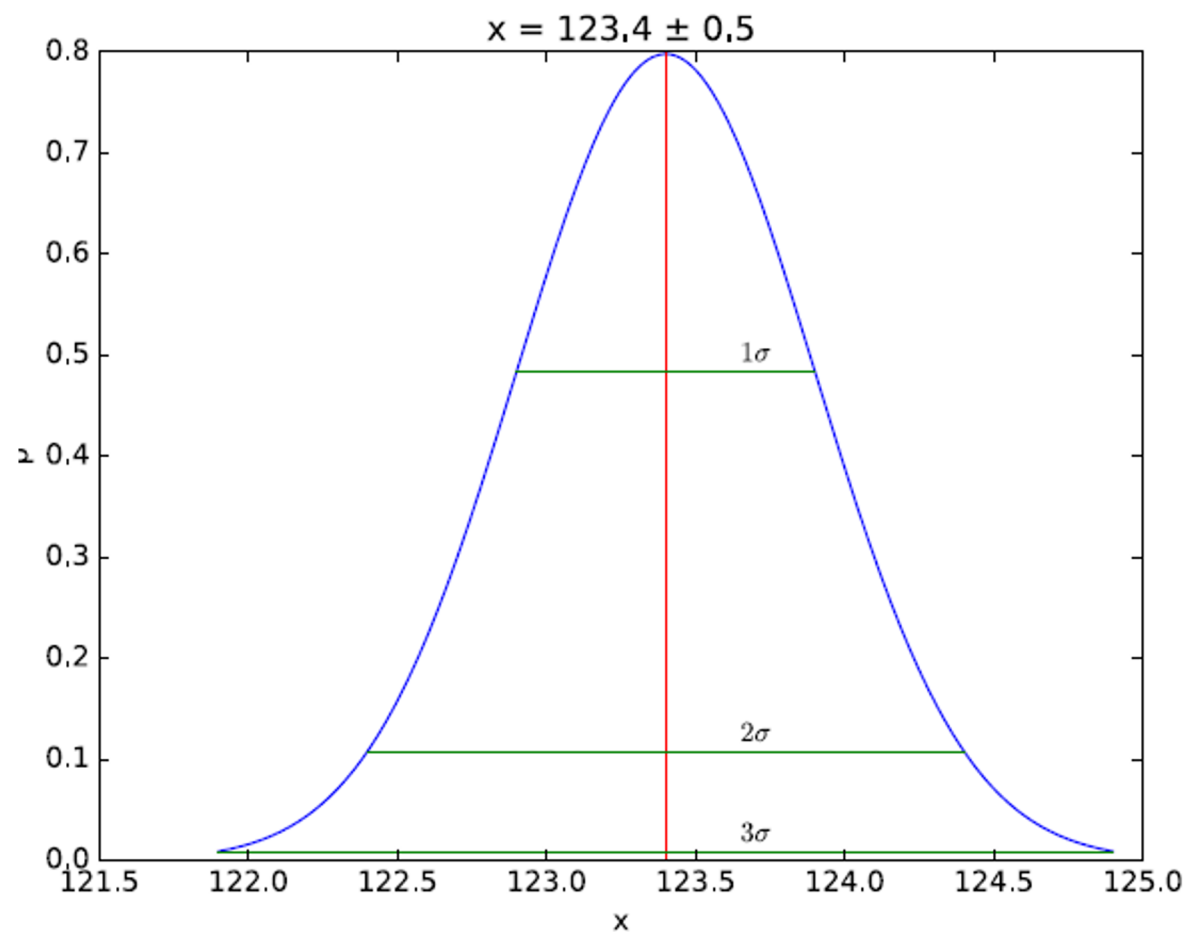
$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = 1$$

mas

$$\int_{-\sigma}^{\sigma} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \approx 0.682$$

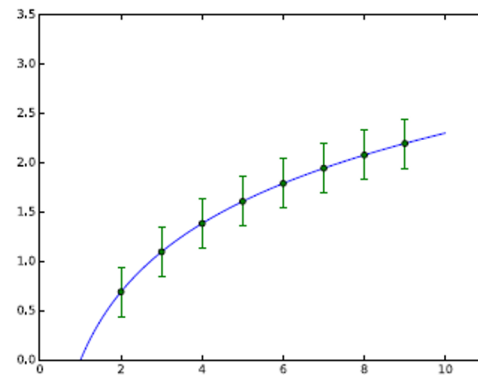
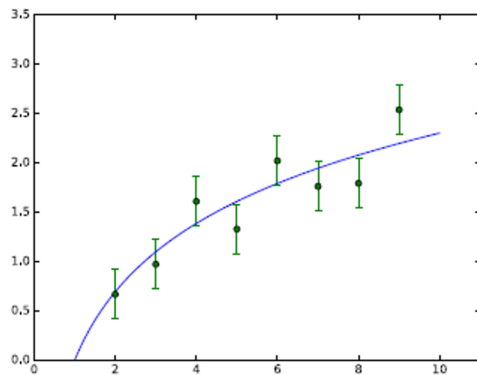
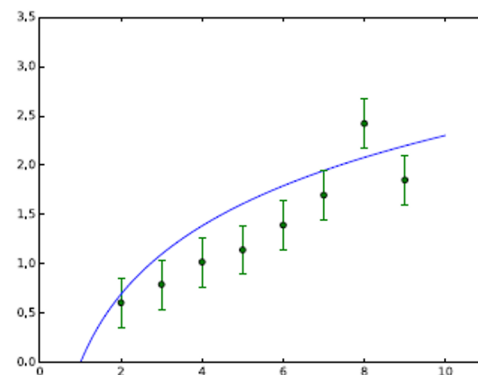
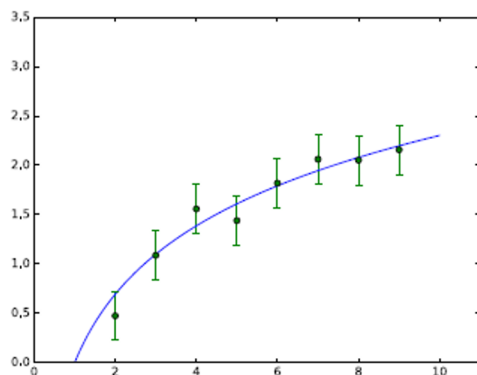
Então só 68.2% dos valores obtidos encontram-se no intervalo
 $x = 123.4 \pm 0.5$!

O significado do erro

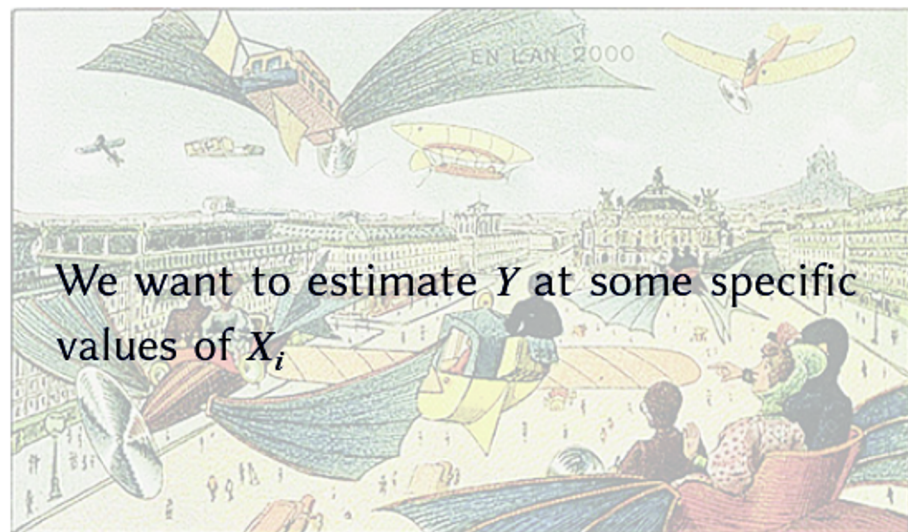


O significado do erro

- Estes dados são compatíveis com a teoria representada por uma curva?

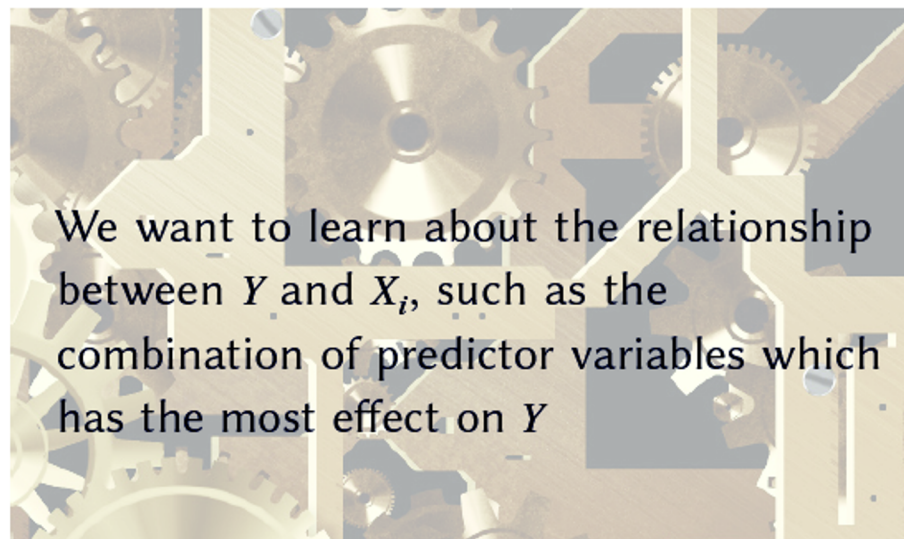


Prediction



We want to estimate Y at some specific values of X_i

Model inference



We want to learn about the relationship between Y and X_i , such as the combination of predictor variables which has the most effect on Y

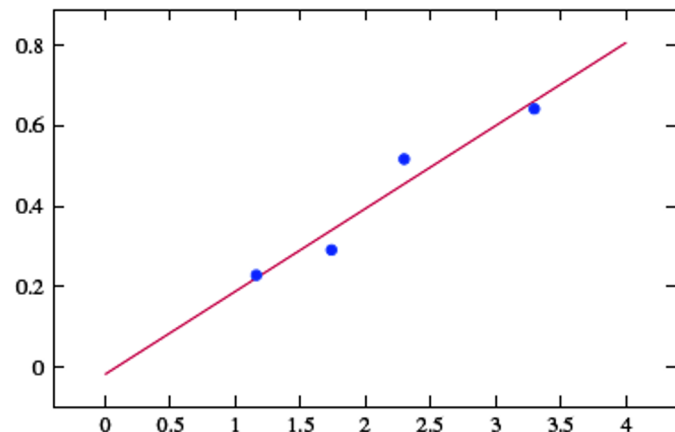
Regressão linear

- ▷ Linear regression: one of the simplest and most commonly used statistical modeling techniques
- ▷ Makes strong assumptions about the relationship between the predictor variables (X_i) and the response (Y)
 - (a linear relationship, a straight line when plotted)
 - only valid for *continuous* outcome variables (not applicable to category outcomes such as *success/failure*)

$$y = \beta_0 + \beta_1 \times x + \text{error}$$

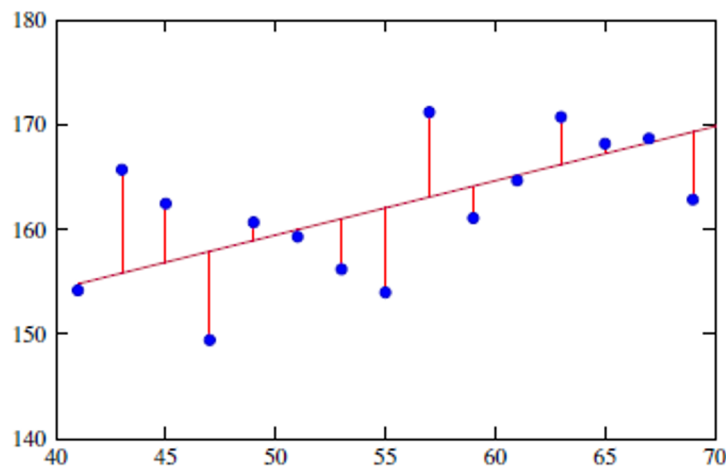
Diagram illustrating the components of the linear regression equation:

- y : outcome variable
- β_0 : intercept
- β_1 : slope
- x : predictor variable

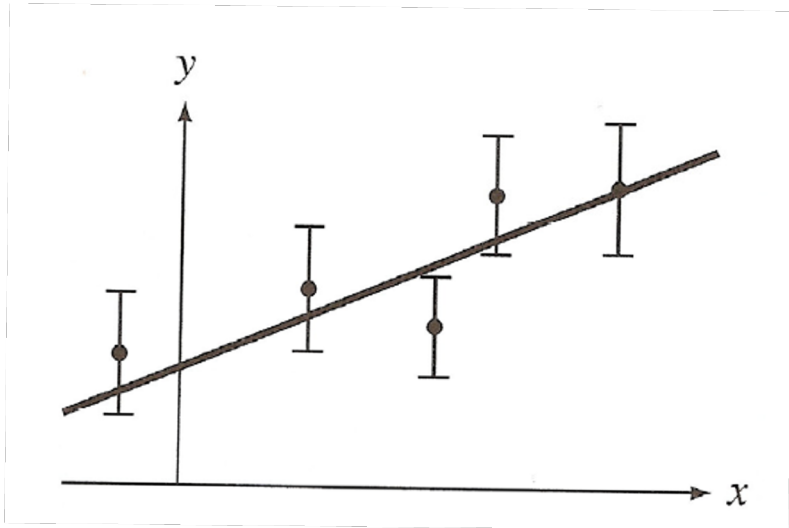


Regressão linear

- ▷ Assumption: $y = \beta_0 + \beta_1 \times x + \text{error}$
- ▷ Our task: estimate β_0 and β_1 based on the available data
- ▷ Resulting model is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \times x$
 - the “hats” on the variables represent the fact that they are **estimated from the available data**
 - $\hat{y}$ is read as “*the estimator for y*”
- ▷ β_0 and β_1 are called the model *parameters* or *coefficients*
- ▷ **Objective:** minimize the *error*, the difference between our observations and the predictions made by our linear model
 - minimize the length of the red lines in the figure to the right (called the “residuals”)



Ajuste por mínimos quadrados



$$\chi^2 = \sum_{i=1}^N \frac{(y_i - A - Bx_i)^2}{\sigma_y^2}$$

$$\frac{\partial \chi^2}{\partial A} = \frac{-2}{\sigma_y^2} \sum_{i=1}^N (y_i - A - Bx_i) = 0$$

$$\frac{\partial \chi^2}{\partial B} = \frac{-2}{\sigma_y^2} \sum_{i=1}^N x_i (y_i - A - Bx_i) = 0$$

Ajuste por mínimos quadrados

$$AN + B \sum x_i = \sum y_i,$$

$$A \sum x_i + B \sum x_i^2 = \sum x_i y_i$$

$$A = \frac{\sum_i x_i^2 \sum_i y_i - \sum_i x_i \sum_i x_i y_i}{\Delta}$$

$$B = \frac{N \sum_i x_i y_i - \sum_i x_i \sum_i y_i}{\Delta},$$

$$\Delta = N \sum_i x_i^2 - \left(\sum_i x_i \right)^2.$$